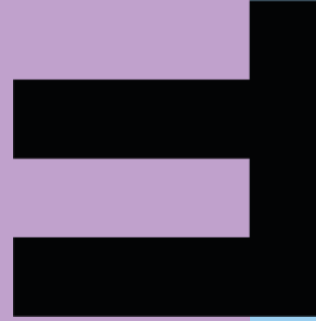


FHV

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of Applied Sciences



Technical Information Systems

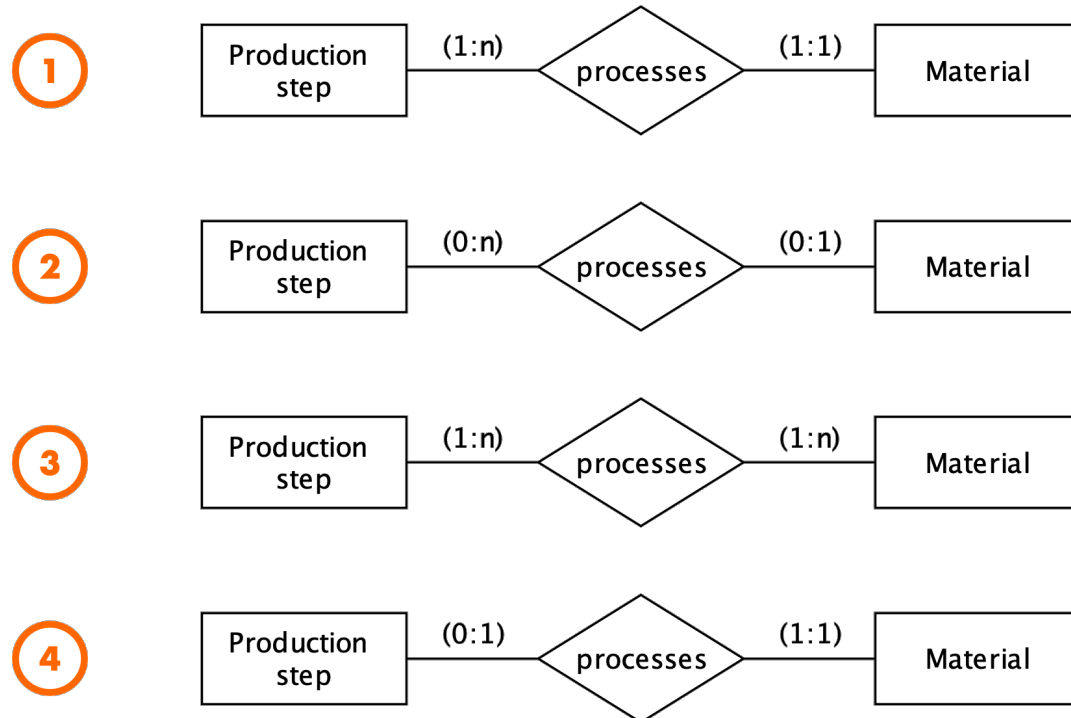
Entity Relationship Modelling – Exercises 1

Exercise

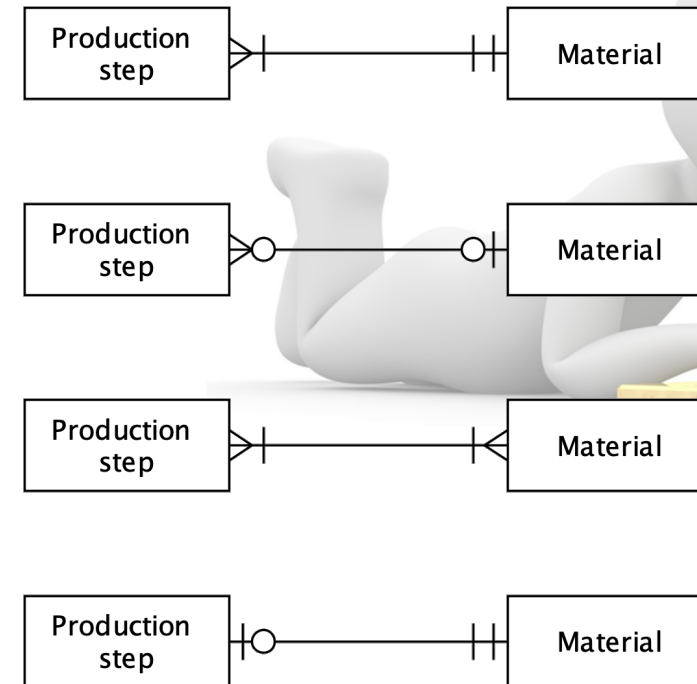
Interpretation of simple models

- Interpret the following models:

Chen (min:max)



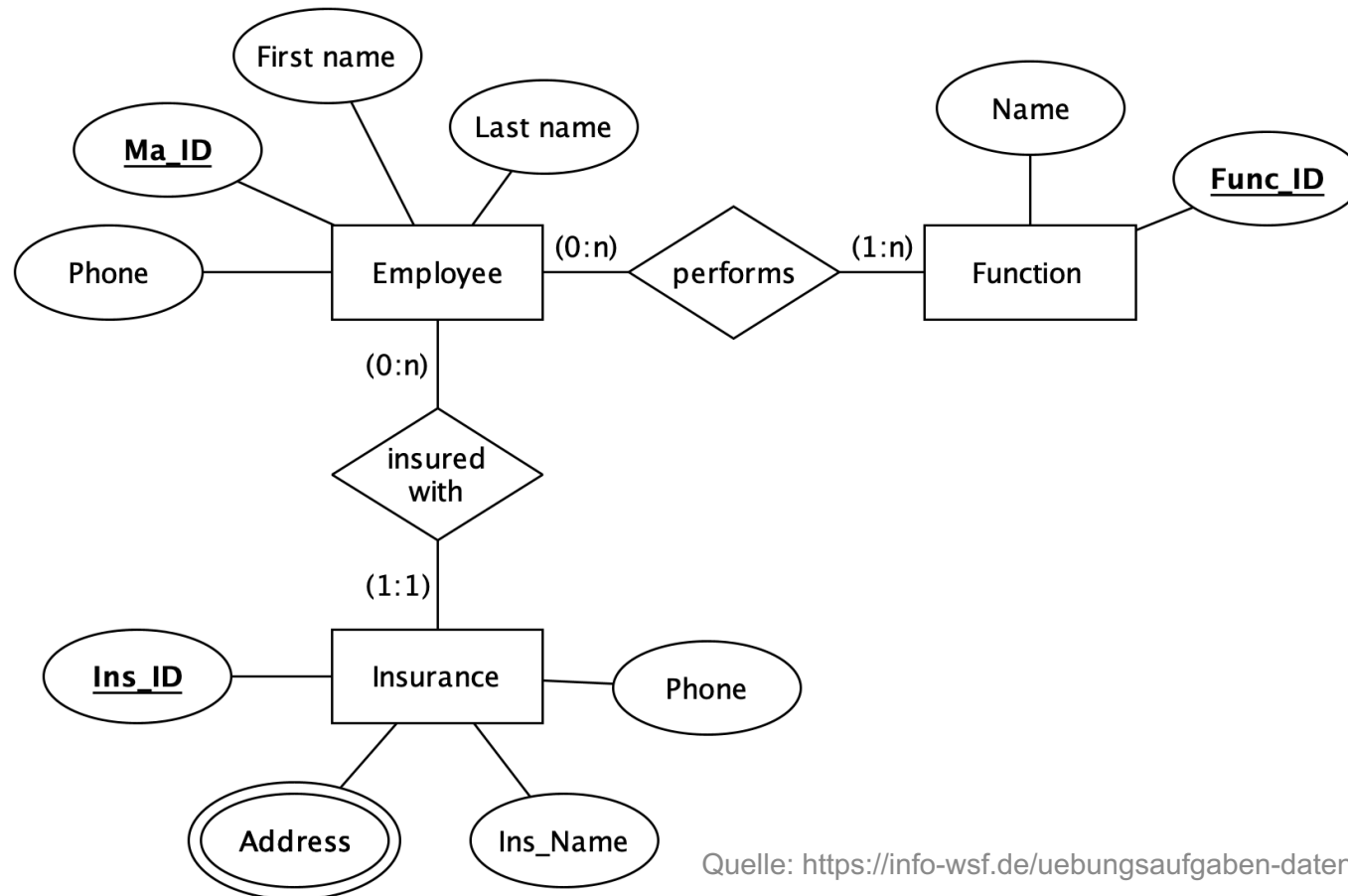
Martin/Crow's foot



Exercise

Interpretation of ER-Models II

- Interpret the following model.



Quelle: https://info-wsf.de/uebungsaufgaben-datenmodellierung/#Aufgabe_4



Exercise

First Modeling Exercise

- For the time being, we are working with pen and paper
- Think about the relationships between the entities people and places.
Create a Chen (with Min-Max) and a Crow-Feet diagram



1. Born
2. Deceased
3. Current Residence
4. Been in
5. Vacation Resorts



- *People* contains the data for any list of people
- You have the information for each person (if available) for the above-mentioned points
- *Places* store data on all places where a person has already been

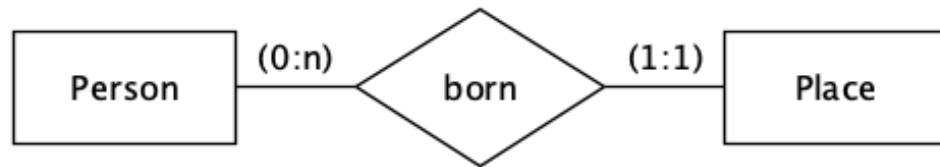
Are there any other assumptions you need to make?



1. Exercise – Solutions 1

Chen (min:max)

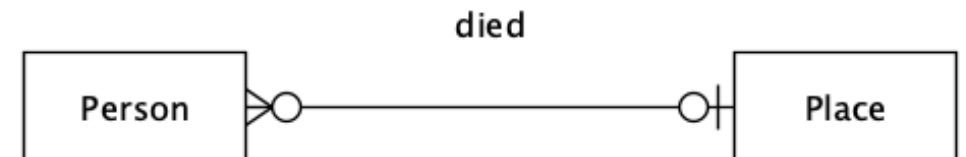
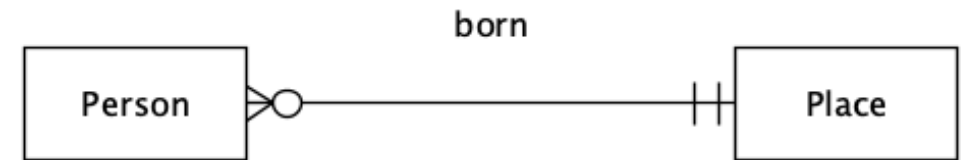
- Person is born in



- Person died in



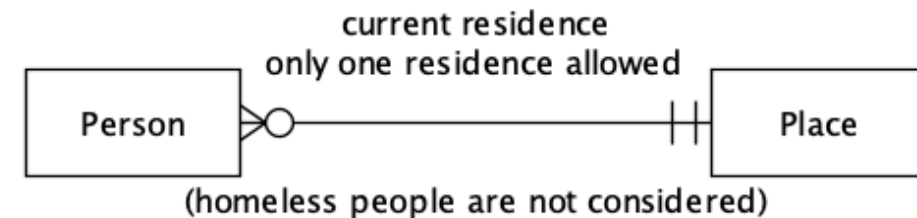
Martin/Crow feet



1. Exercise – Solutions 2a

Current Residence

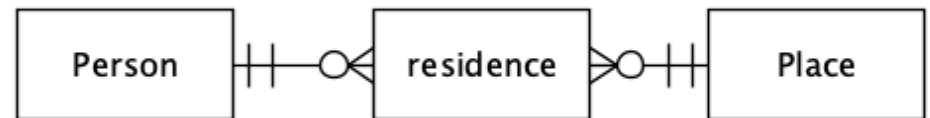
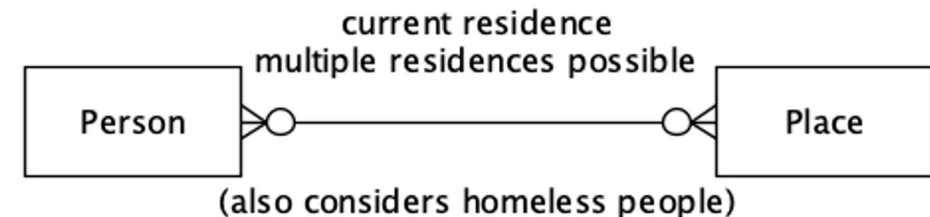
- What do we need to consider?
 - Several people can live in one place
 - Decision: Is it possible to have more than one residence? → NO
 - Decision: Are homeless persons also taken into account? → NO



1. Exercise – Solutions 2b

Current Residence

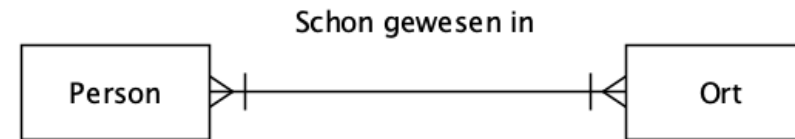
- What do we need to consider?
 - Several people can live in one place
 - Decision: Is it possible to have more than one residence? → YES
 - Decision: Are homeless persons also taken into account? → YES



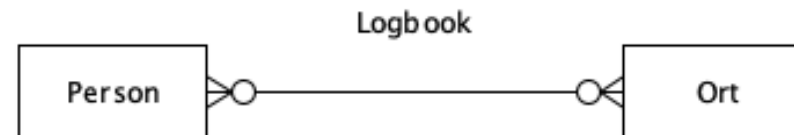
1. Exercise – Solutions 3

Been to

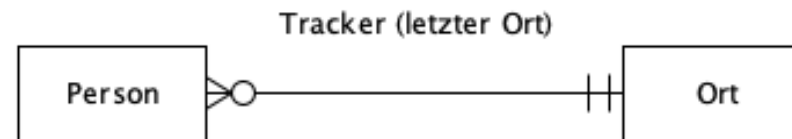
- Logging of all the person's attendances



- Logging of some of the person's presences (logbook)



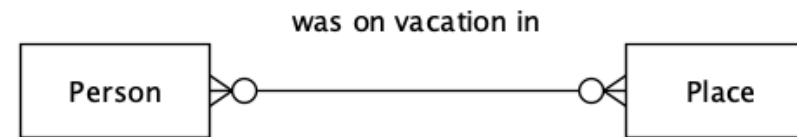
- Tracker (last presence of the person)



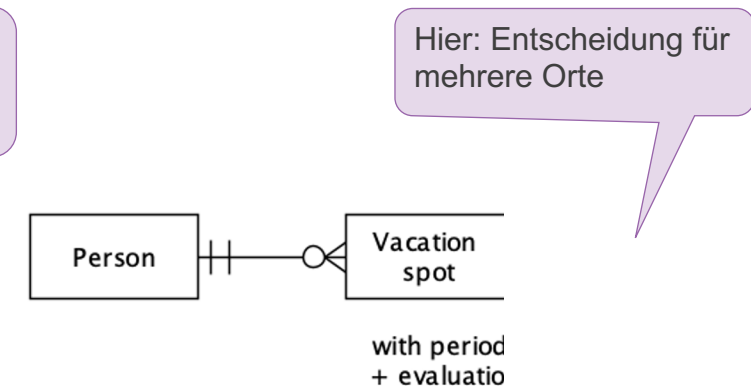
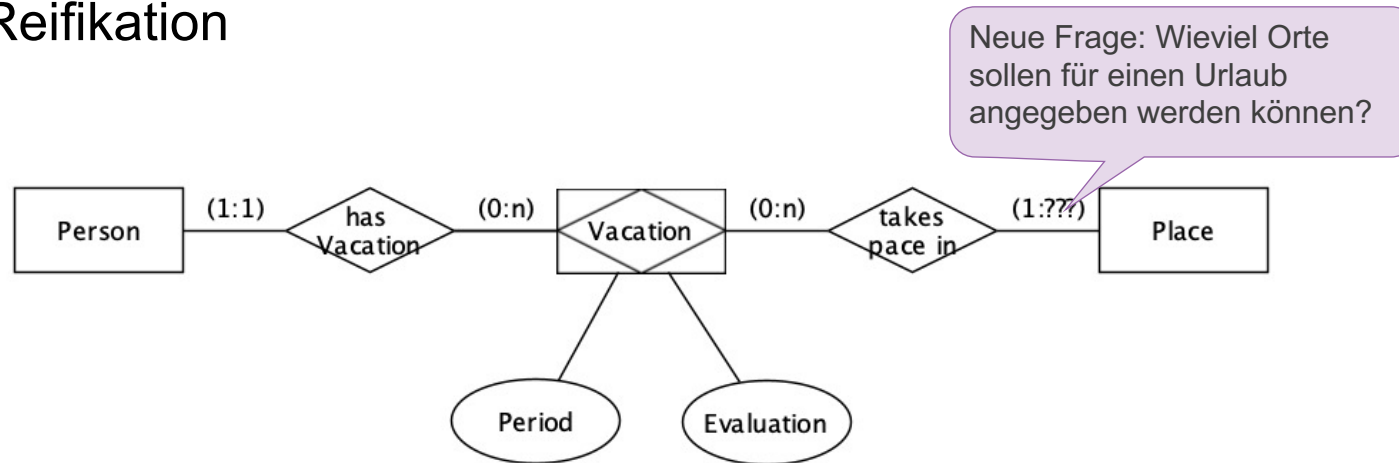
1. Exercise – Solutions 4

Urlaubsort

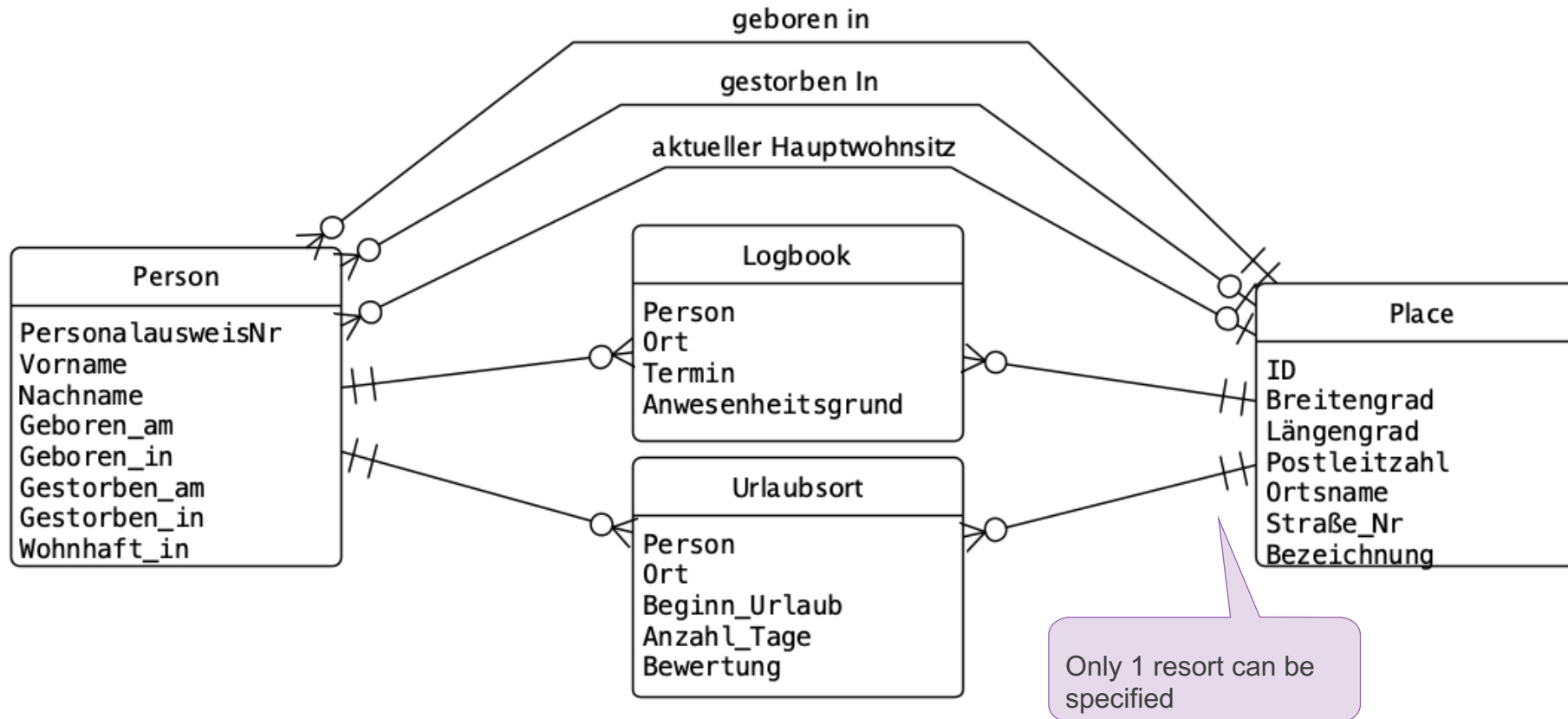
- Alle/mehrere Urlaube der Person werden aufgezeichnet



- Reifikation



1. Exercise – Solutions 5 – all 5 tasks with exemplary attributes



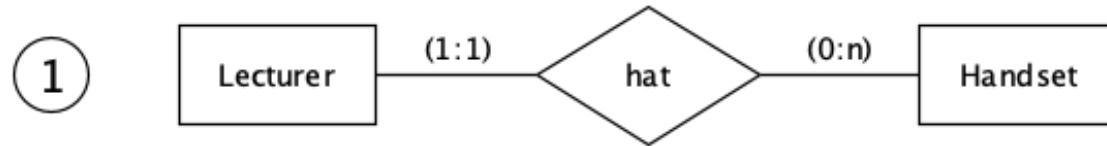
2. Library data

- Model as Chen (min-max) and as Crow-Feet (each task separately):
 1. Lecturer has "handset". A book is ordered for him/her and remains in his/her possession permanently.
 2. Currently loaned books to persons (i.e. only the information person has a book)
 3. Extension of 2: A book title can be borrowed in several copies. Copies may also be blocked for borrowing (reference copies).
 4. Extension of 3.: The return can be extended twice if the book has not been reserved (more often than twice in exceptional cases).
 5. Extension of 4: Borrowing with history (transactions of books that have already been returned are also saved)
- Finally, solution 5 with attributes (some examples)

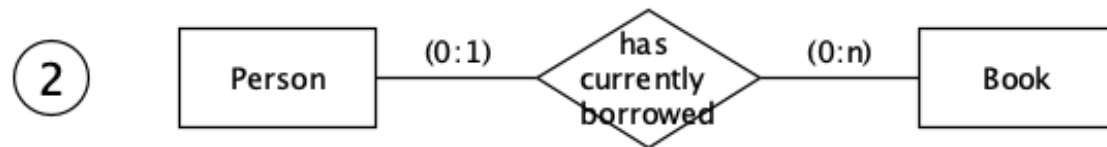


2. Library data – Solutions 1

1. Lecturer has "handset" (A book is ordered for him/her and remains permanently in his/her possession. → List of books)

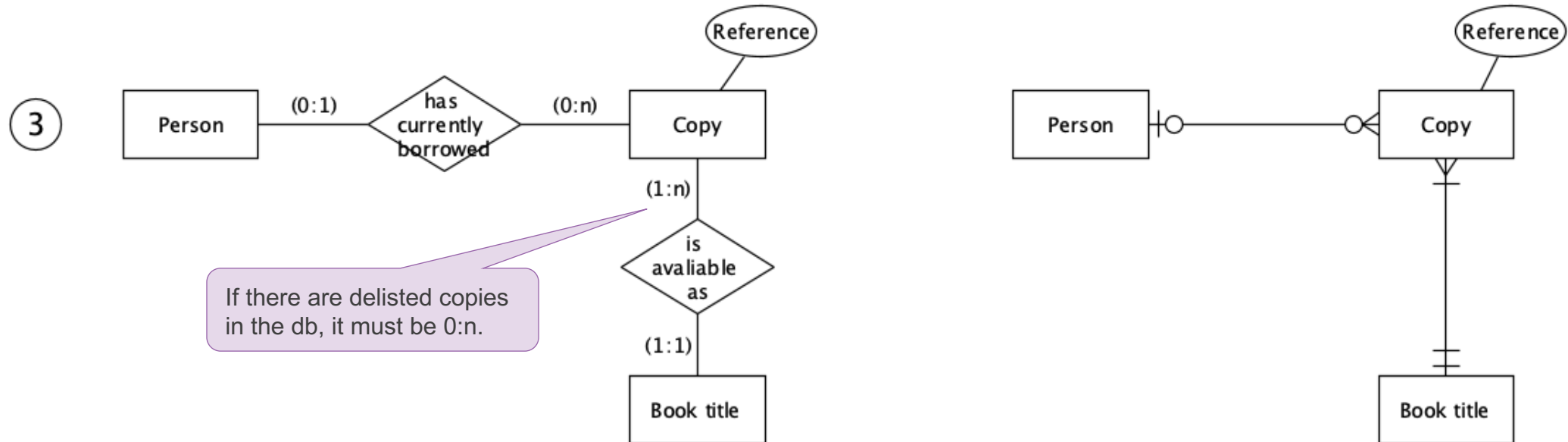


2. Currently loaned books to persons (i.e. only the information person has book)



2. Library data – Solutions 2

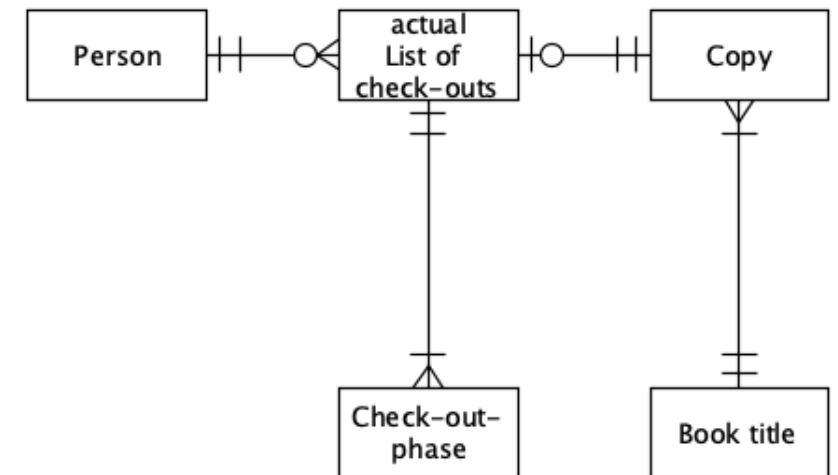
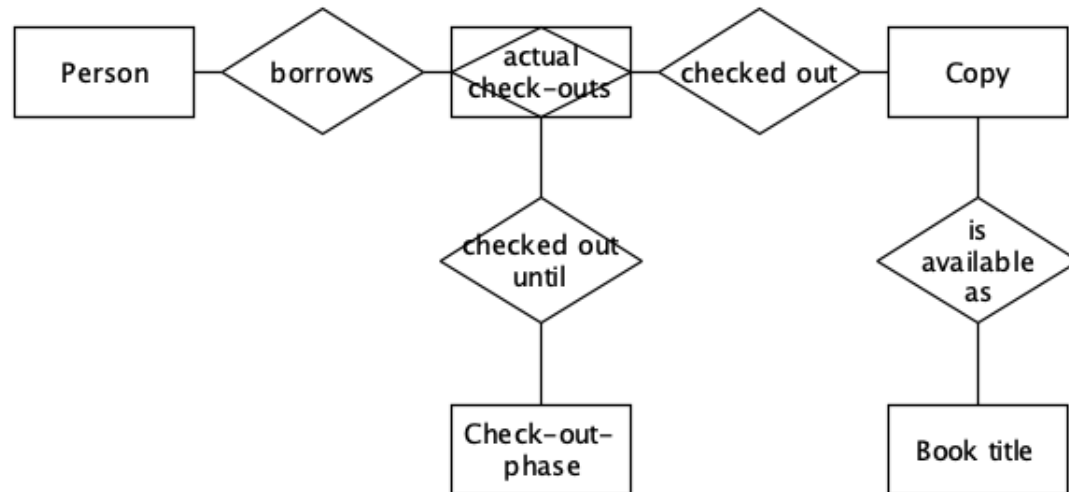
3. Extension of 2: A book title can be borrowed in several copies. Copies may also be blocked for borrowing (reference copies).



2. Library data – Solutions 3

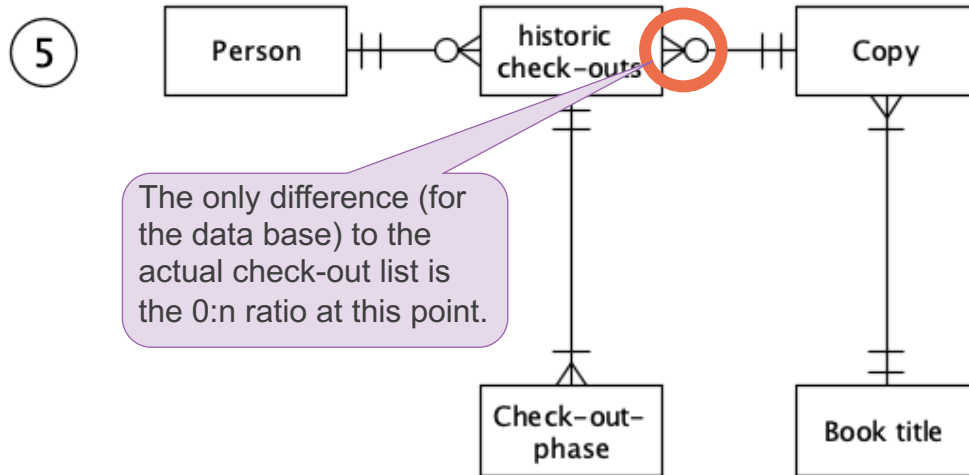
4. Extension of 3.: The return can be extended twice if the book has not been reserved (more often in exceptional cases).

4



2. Library data – Solutions 3

5. Extension of 4: Borrowing with history (transactions of books that have already been returned are also saved)



2. Library data – Solutions 4 with some attributes (logical model)

